

NICHOLAS ROBERTS

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EDUCATION

University of Wisconsin - Madison

August 2021 - Present

Ph.D. Computer Science, Mathematics minor

- Advisor: Frederic Sala

Carnegie Mellon University

August 2019 - May 2021

M.S. Machine Learning

- Advisors: Ameet Talwalkar, Zachary C. Lipton

University of California San Diego

September 2015 - March 2019

B.S. Computer Science, Mathematics minor

- Advisors: Sanjoy Dasgupta, Garrison W. Cottrell

- Magna Cum Laude with CSE department Highest Distinction honors

Fresno City College

August 2013 - May 2015

PUBLICATIONS

*equal contribution, +alphabetical

CONFERENCE PUBLICATIONS

Nicholas Roberts, Samuel Guo, Zhiqi Gao, Satya Sai Srinath Namburi GNVV, Sonia Crompt, Chengjun Wu, Chengyu Duan, Frederic Sala. (2025). *Pretrained Hybrids with MAD Skills*. Conference on Language Modeling (COLM) 2025.

Deepak Nathani, Lovish Madaan, **Nicholas Roberts**, Nikolay Bashlykov, Ajay Menon, Vincent Moens, Amar Budhiraja, Despoina Magka, Vladislav Vorotilov, Gaurav Chaurasia, Dieuwke Hupkes, Ricardo Silveira Cabral, Tatiana Shavrina, Jakob Foerster, Yoram Bachrach, William Yang Wang, Roberta Raileanu. (2025). *MLGym: A New Framework and Benchmark for Advancing AI Research Agents*. Conference on Language Modeling (COLM) 2025.

Nicholas Roberts, Niladri Chatterji, Sharan Narang, Mike Lewis, Dieuwke Hupkes. (2025). *Compute Optimal Scaling of Skills: Knowledge vs Reasoning*. Association for Computational Linguistics (ACL) Findings, 2025.

Tianyi Zhang*, Linrong Cai*, Jeffrey Li, **Nicholas Roberts**, Neel Guha, Frederic Sala. (2024). *Stronger Than You Think: Benchmarking Weak Supervision on Realistic Tasks*. Neural Information Processing Systems (NeurIPS) Datasets and Benchmarks Track, 2024.

Nicholas Roberts, Xintong Li, Dyah Adila, Sonia Crompt, Tzu-Heng Huang, Jitian Zhao, Frederic Sala. (2023). *Geometry-Aware Adaptation for Pretrained Models*. Neural Information Processing Systems (NeurIPS), 2023.

Mayee Chen, **Nicholas Roberts**, Kush Bhatia, Jue Wang, Ce Zhang, Frederic Sala, Christopher Ré. (2023). *Skill-it! A data-driven skills framework for understanding and training language models*. Neural Information Processing Systems (NeurIPS), 2023 (spotlight).

Benedikt Boecking, **Nicholas Roberts**, Willie Neiswanger, Stefano Ermon, Frederic Sala, Artur Dubrawski. (2023). *Generative Modeling Helps Weak Supervision (and Vice Versa)*. International Conference on Learning Representations (ICLR), 2023.

Nicholas Roberts*, Xintong Li*, Tzu-Heng Huang, Dyah Adila, Spencer Schoenberg, Cheng-Yu Liu, Lauren Pick, Haotian Ma, Aws Albarghouthi, Frederic Sala. (2022). *AutoWS-Bench-101: Benchmarking Automated Weak Supervision with 100 Labels*. Neural Information Processing Systems (NeurIPS) Datasets and Benchmarks Track, 2022.

Renbo Tu*, **Nicholas Roberts***, Mikhail Khodak, Junhong Shen, Frederic Sala, Ameet Talwalkar. (2022). *NAS-Bench-360: Benchmarking Neural Architecture Search on Diverse Tasks*. Neural Information Processing Systems (NeurIPS) Datasets and Benchmarks Track, 2022.

Harit Vishwakarma, **Nicholas Roberts**, Frederic Sala. (2022). *Lifting Weak Supervision To Structured Prediction*. Neural Information Processing Systems (NeurIPS), 2022.

Changho Shin, Winfred Li, Harit Vishwakarma, **Nicholas Roberts**, Frederic Sala. (2022). *Universalizing Weak Supervision*. International Conference on Learning Representations (ICLR), 2022.

Nicholas Roberts*, Mikhail Khodak*, Tri Dao, Liam Li, Christopher Ré, Ameet Talwalkar. (2021). *Rethinking Neural Operations for Diverse Tasks*. Neural Information Processing Systems (NeurIPS), 2021.

Sanjoy Dasgupta, Akansha Dey, **Nicholas Roberts**, Sivan Sabato. (2018). *Learning from discriminative feature feedback*. Neural Information Processing Systems (NeurIPS), 2018.

JOURNAL PUBLICATIONS

Sonia Crompt, Satya Sai Srinath Namburi GNVV, Catherine Cao, Mohammed Alkhudhayri, Samuel Guo, **Nicholas Roberts**, Frederic Sala. (2025). *Tabby: Tabular Adaptation for Language Models*. Transactions on Machine Learning Research (TMLR) 2025.

Nicholas Roberts*, Samuel Guo*, Cong Xu*, Ameet Talwalkar, David Lander, Lvfang Tao, Linhang Cai, Shuaicheng Niu, Jianyu Heng, Hongyang Qin, Minwen Deng, Johannes Hog, Alexander Pfefferle, Sushil Amanaghatta Shivakumar, Arjun Krishnakumar, Yubo Wang, Rhea Sukthanker, Frank Hutter, Euxhen Hasanaj, Tien-Dung Le, Mikhail Khodak, Yuriy Nevmyvaka, Kashif Rasul, Frederic Sala, Anderson Schneider, Junhong Shen, Evan Sparks. (2023). *AutoML Decathlon: Diverse Tasks, Modern Methods, and Efficiency at Scale*. NeurIPS 2022 Competition Track, Proceedings of Machine Learning Research (PMLR).

Aarohi Srivastava⁺, ..., **Nicholas Roberts**⁺ (276), ..., (442 authors). (2023). *Beyond the Imitation Game: Quantifying and Extrapolating the Capabilities of Language Models*. Transactions on Machine Learning Research (TMLR) 2023 (Finalist for Outstanding Certification) and ICLR 2025.

Kaustubh D. Dhole, ..., **Nicholas Roberts** (85), ..., (128 authors). (2023). *NL-Augmenter: A Framework for Task-Sensitive Natural Language Augmentation*. Northern European Journal of Language Technology (NEJLT) 2023.

Chen Zhang*, Yerlan Idelbayev*, **Nicholas Roberts**, Yiwen Tao, Yashwanth Nannapaneni, Brendan M. Duggan, Jie Min, Eugene C. Lin, Erik C. Gerwick, Garrison W. Cottrell, William H. Gerwick. (2017). *Small Molecule Accurate Recognition Technology (SMART) to Enhance Natural Products Research*. Scientific Reports.

WORKSHOP PUBLICATIONS & PREPRINTS

Nicholas Roberts, Sungjun Cho, Zhiqi Gao, Tzu-Heng Huang, Albert Wu, Gabriel Orlanski, Avi Trost, Kelly Buchanan, Aws Albarghouthi, Frederic Sala. (2026). *Test-Time Scaling Makes Overtraining Compute-Optimal*. Preprint.

Yijing Zhang, **Nicholas Roberts**, Tanya Marwah, Mikhail Khodak. (2026). *Breakeven complexity: A new perspective on evaluating neural partial differential equation solvers*. Preprint.

Gabriel Orlanski, **Nicholas Roberts**, Aws Albarghouthi, Frederic Sala. (2026). *Pareto Optimal Code Generation*. Preprint.

Alexander Goldie, Zilin Wang, Adrian Hayler, Deepak Nathani, Edan Toledo, Aleksandra Kalisz, Ken Thampirawong, Michael Beukman, Alistair Letcher, Shashank Reddy Chirra, Clarisse Wibault, Theo Wolf, Charles O'Neill, Uljad Berdica, **Nicholas Roberts**, Saeed Rahmani, Roberta Raileanu, Shimon Whiteson, Jakob Foerster. (2026). *Procedural Generation Of Algorithm Discovery Tasks in Machine Learning*. Preprint.

Albert Ge, Tzu-Heng Huang, John Cooper, Avi Trost, Ziyi Chu, Satya Sai Srinath Namburi GNVV, Ziyang Cai, Kendall Park, **Nicholas Roberts**, Frederic Sala. (2025). *R&B: Domain Regrouping and Data Mixture Balancing for Efficient Foundation Model Training*. ICML 2025 DIG-BUGS Workshop. ICML 2025 DataWorld Workshop.

Wenxuan Tan, **Nicholas Roberts**, Tzu-Heng Huang, Jitian Zhao, John Cooper, Samuel Guo, Chengyu Duan, Frederic Sala. (2024). *MoRe Fine-Tuning with 10x Fewer Parameters*. ICML 2024 Efficient Systems for Foundation Models (ES-FoMo) Workshop. ICML 2024 Workshop on Foundation Models in the Wild.

Nicholas Roberts, Yingyu Liang, Frederic Sala. (2023). *Understanding Neural Architecture Search by its Architecture Parameters*. Midwest Machine Learning Symposium 2023.

Tzu-Heng Huang, Harit Vishwakarma, Catherine Cao, Spencer Schoenberg, **Nicholas Roberts**, Frederic Sala. (2023). *ScriptoriumWS: A Code Generation Assistant for Weak Supervision*. ICLR 2023 Deep Learning for Code Workshop.

Renbo Tu, **Nicholas Roberts**, Vishak Prasad, Sibasis Nayak, Paarth Jain, Frederic Sala, Ganesh Ramakrishnan, Ameet Talwalkar, Willie Neiswanger, Colin White. (2022). *AutoML for Climate Change: A Call to Action*. NeurIPS 2022 Tackling Climate Change with Machine Learning Workshop.

Nicholas Roberts, Davis Liang, Graham Neubig, Zachary C. Lipton. (2020). *Decoding and Diversity in Machine Translation*. NeurIPS 2020 Resistance AI Workshop.

Mikhail Khodak, Liam Li, **Nicholas Roberts**, Maria-Florina Balcan, Ameet Talwalkar. (2020). *A Simple Setting for Understanding Neural Architecture Search with Weight-Sharing*. ICML 2020 AutoML Workshop.

Mikhail Khodak*, Liam Li*, **Nicholas Roberts**, Maria-Florina Balcan, Ameet Talwalkar. (2020). *Weight-Sharing Beyond Neural Architecture Search: Efficient Feature Map Selection and Federated Hyperparameter Tuning*. MLSys 2020 On-Device Intelligence Workshop.

Nicholas Roberts, Dian A. Yap, Vinay U. Prabhu. (2019). *Deep Connectomics Networks: Neural Network Architectures Inspired by Neuronal Networks*. NeurIPS 2019 Real Neurons and Hidden Units Workshop.

Nicholas Roberts, Poornav S. Purushothama, Vishal T. Vasudevan, Siddarth Ravichandran, Chen Zhang, William H. Gerwick, Garrison W. Cottrell. (2019). *Using Deep Siamese Neural Networks to Speed up Natural Products Research*. NeurIPS 2019 workshop on Machine Learning and the Physical Sciences.

Dian A. Yap, **Nicholas Roberts**, Vinay U. Prabhu. (2019). *Grassmannian Packings in Neural Networks: Learning with Maximal Subspace Packings for Diversity and Anti-Sparsity*. NeurIPS 2019 Workshop on Bayesian Deep Learning.

Nicholas Roberts, Vinay U. Prabhu, Matthew McAteer. (2019). *Model Weight Theft With Just Noise Inputs: The Curious Case of the Petulant Attacker*. ICML 2019 Workshop on Security and Privacy of Machine Learning (spotlight).

PRESENTATIONS

Compute Optimal Scaling of Skills: Knowledge vs Reasoning

Invited Talk Data Foundations of AI Seminar.

Host: Tzu-Heng Huang

Online. March 2026.

Invited Talk Jingbo Shang’s group.

Hosts: Xintong Li, Jingbo Shang

UCSD – La Jolla, California. November 2025.

Invited Talk Princeton Language and Intelligence Seminar.

Host: Xingyu Zhu

Princeton University – Princeton, New Jersey. October 2025.

Invited Talk Google ML Foundations Seminar.

Host: Zak Mhammedi

Google – New York, New York. September 2025.

Invited Talk Azalia Mirhoseini’s group.

Host: Azalia Mirhoseini

Stanford University – Stanford, California. August 2025.

Invited Talk Christopher Ré’s group.

Host: Christopher Ré

Stanford University – Stanford, California. August 2025.

Invited Talk Sanmi Koyejo’s group.

Host: Sanmi Koyejo

Stanford University – Stanford, California. August 2025.

Invited Talk Cerebras Research Seminar.

Hosts: Nolan Dey, Joel Hestness

Cerebras – Sunnyvale, California. August 2025.

Invited Talk Microsoft Research New England.

Host: David Alvarez-Melis

Online. August 2025.

Invited Talk Ludwig Schmidt’s group.

Host: Ludwig Schmidt

Stanford University – Stanford, California. August 2025.

Invited Talk Ploutos.

Host: Cecilia Tamura

Online. May 2025.

Invited Talk William Yang Wang’s group.

Host: Deepak Nathani

UCSB – Santa Barbara, California. March 2025.

Invited Talk Meta Generative AI Tech Talks.

London, England. February 2025.

Hybrid Foundation Models

Guest Lecture UW-Madison CS 839: Foundation Models.

Madison, WI. October 2024.

Geometry-Aware Adaptation for Pretrained Models

Invited Talk CMU AI Seminar.

Hosts: Zico Kolter, Asher Trockman

Carnegie Mellon University – Pittsburgh, PA. October 2023.

Geometry-Aware Adaptation for Pretrained Models

Invited Talk UW Madison IFDS Ideas Forum.

Host: Sebastien Roch

Madison, WI. October 2023.

AutoML Cup 2023

Invited Talk AutoML Conference 2023.

Host: Frank Hutter

Hasso-Plattner Institut – Potsdam/Berlin, Germany. September 2023.

Toward Data-Structured Prediction

Lightning Talk MLCommons Rising Stars 2023 Workshop.

Hosts: Udit Gupta, Abdulrahman Mahmoud, Lillian Pentecost

Google – Sunnyvale, CA. August 2023.

Geometry-Aware Adaptation for Pretrained Models

Invited Talk Microsoft Research, ML Foundations Seminar.

Host: Sébastien Bubeck.

Microsoft – Redmond, WA. August 2023.

The AutoML Decathlon: Diverse Tasks, Modern Methods, and Efficiency at Scale

Invited Talk AutoML Seminar.

Host: Aaron Klein.

Online. April 2023.

AutoML Decathlon Hackathon

Invited Hackathon AutoML Fall School.

Host: Frank Hutter.

Freiburg, Germany. October 2022.

Rethinking AutoML for Diverse Tasks

Invited Talk Physics $\cap$ ML Seminar, University of Wisconsin - Madison.
Host: Gary Shiu
Madison, WI. March 2022.

Searching for Convolutions and a More Ambitious NAS

Plenary Talk AAAI 2021 Workshop on Learning Network Architecture During Training.
Online. February 2021.

Model Weight Theft With Just Noise Inputs: The Curious Case of the Petulant Attacker

Spotlight Presentation ICML 2019 Workshop on Security and Privacy of Machine Learning.
Long Beach, CA, USA. June 2019.

Small Molecule Accurate Recognition Technology: A Digital Frontier to Reshape Natural Product Research

Spotlight Presentation Applied Machine Learning Days 2018.
EPFL – Lausanne, Switzerland. January 2018.

AWARDS

Jane Street Ph.D. Fellowship (honorable mention) <i>Jane Street Capital</i>	2025
NeurIPS Scholar Award <i>Neural Information Processing Systems (NeurIPS)</i>	2023
ML and Systems Rising Stars Award <i>MLCommons</i>	2023
NeurIPS Scholar Award <i>Neural Information Processing Systems (NeurIPS)</i>	2022
Prove AI Fellowship <i>Prove AI Labs</i>	2021
First-Year CS Departmental Scholarship <i>University of Wisconsin - Madison</i>	2021
NeurIPS “Travel” Award <i>Neural Information Processing Systems (NeurIPS)</i>	2020
UnifyID AI Fellowship <i>UnifyID AI Labs</i>	2019
Outstanding Undergraduate Researcher Award (honorable mention) <i>Computing Research Association (CRA)</i>	2019
NeurIPS Travel Award <i>Neural Information Processing Systems (NeurIPS)</i>	2018
Best Spotlight Presentation Award <i>Applied Machine Learning Days (AMLD)</i>	2018

EXPERIENCE

University of Wisconsin - Madison <i>Research Assistant (Sprocket Lab led by Frederic Sala)</i>	August 2021 -
· Ph.D. research on Foundation Models, Data-Centric AI, and scientific applications advised by Frederic Sala	
Meta <i>Research Scientist Intern (Llama Generative AI pretraining with Dieuwke Hupkes)</i>	August 2024 - February 2025
· Research on skills and scaling laws for LLM training.	

- Together AI** May 2024 - August 2024
Research Intern (mentored by Tri Dao)
- Research on mechanistic interpretability of hybrid LLM architectures.
- Microsoft Research** June 2023 - September 2023
Research Intern (Physics of AGI group led by Sébastien Bubeck)
- Developed activation function search techniques for large-scale LLM pretraining
 - Developed learning curve extrapolation techniques to ablate architectural choices in transformers
- Carnegie Mellon University** May 2020 - August 2020, May 2021 - August 2021
Research Assistant (SAGE Lab led by Ameet Talwalkar)
- Explored two directions for expanding NAS search spaces: large scale edge learning and operation learning
 - Gave monthly research presentations to J.P. Morgan researchers
- Amazon AWS AI** June 2019 - August 2019
Applied Scientist Intern (AWS Transcribe research led by Katrin Kirchhoff)
- Identified areas for improvement in existing ASR systems when recognizing rare or zero shot entities
 - Researched and developed methods for hypothesis rescoring in ASR systems using neural language modeling
- UnifyID** February 2019 - June 2019
Machine Learner Intern (mentored by Vinay Uday Prabhu)
- Developed a novel model extraction attack against deep learning models for computer vision using just noise inputs
 - Researched ways to apply network neuroscience findings to deep learning
- Intuit** June 2018 - September 2018
Software Engineering Intern
- Researched and implemented a novel controllable text generation model as a service within Intuit
 - Identified dynamic topic models as a promising direction for analyzing customer support tickets over time
- Altum** January 2018 - May 2018
Applied Scientist Intern
- Developed language model to extract NLP features from text data for cryptocurrency trading
 - Implemented SoTA unsupervised sentiment analysis models for classifying streaming online forum data
- UCSD CSE Department** September 2017 - March 2018
Data Science Tutor
- Tutored DSC 10 Introduction to Data Science, under Professor Janine Tiefenbruck
 - Tutored DSC 20 Principles of Data Science, under Professor Marina Langlois
- Teradata** June 2017 - September 2017
Software Engineering Intern
- Improved training methodology and architecture of deep learning time series model used internally
 - Developed open source Spark-Teradata connector forked from Databricks connector for AWS Redshift
- Skqrl** June 2016 - December 2016
Software Engineering Intern
- Developed web scraping tool to compile product data
 - Designed and implemented search pipeline and database using Python, Django, and MySQL
- ModSpot** January 2016 - March 2016
Software Engineering Intern
- Implemented new user account, edit profile, and login designs in Objective-C for iOS application
 - Refactored analytics code for gathering statistics on app usage
- The Comeback Community** June 2015 - September 2015
Volunteer Full Stack Developer

- Developed website for educational nonprofit using Google Cloud Platform

Fresno City College

January 2015 - May 2015

Tutor

- Tutored calculus, linear algebra, data structures, discrete mathematics, and Android app development

Fresno County Sheriff's Office

May 2013 - August 2013

IT Intern

- Replaced malfunctioning hardware in employee PCs

FUNDING AWARDED

DARPA

2025 - 2026

- Advanced Research Concepts (ARC) SAFRON Award. "ProD SAFE: Programmatic Distillation for Safe and Assured Foundation Models & Robots" (Co-PI; PI: Frederic Sala). \$137,998.

MENTORING

Albert Wu

2025 -

- Current position: undergraduate in Computer Science at Wisconsin

Frank Xiang

2025

- Current position: undergraduate in Computer Science at Wisconsin

Kashish Agarwal

2025

- Current position: undergraduate in Computer Science at Wisconsin

Mihir Sahu

2025

- Current position: undergraduate in Computer Science at Wisconsin

Linrong Cai

2023 - 2024

- Current position: M.S. student in Computer Science at Princeton

Tianyi Zhang

2023 - 2024

- Current position: M.S. student in Computer Science at University of Washington

Chengyu Duan

2023 - 2024

- Current position: M.S. student in Computer Science at Georgia Tech

Chengjun Wu

2023 - 2024

- Current position: undergraduate in Computer Science at Wisconsin

Zhiqi Gao

2023 - 2024

- Current position: Ph.D. student in Computer Science at Wisconsin

Wenxuan Tan

2023 - 2024

- Current position: undergraduate in Computer Science at Wisconsin

Spencer Schoenberg

2021 - 2022

- Current position: M.S. student in Computer Science at Wisconsin

Xintong Li

2021 - 2023

- Current position: Ph.D. student in Computer Science at UCSD

LEADERSHIP, ACTIVITIES, SERVICE, AND EXTRACURRICULAR

Wisconsin:	AutoML Conference 2023 NeurIPS 2022 High School Outreach Lead competition organizer: AutoML Cup 2023 AutoML Decathlon competition @ NeurIPS 2022 AutoML Decathlon Hackathon @ Wisconsin AutoML Decathlon Hackathon @ the AutoML Fall School 2022 Reviewer: NeurIPS 2022, 2023, 2024 NeurIPS Datasets and Benchmarks Track 2021, 2022, 2023, 2024, 2025 ICLR 2024, 2025 ICML 2022, 2023, 2024 ICML Efficient Systems for Foundation Models Workshop 2023, 2024, 2025 ICML Long Context Foundation Models Workshop 2025 ICML Mechanistic Interpretability Workshop 2024 AAAI Spring Symposium on Clinical Foundation Models 2024 AISTATS 2023 AutoML 2023, 2025 AutoML 2023 Workshop Track AutoML 2024, 2025 ABCD Track TMLR LoG 2025 Research to Impact (R2I) Scratch Club Wisconsin Hoofer Sailing Club Wisconsin Triathlon Team	Competition Chair Group Leader
CMU:	MSML Student Committee 2019-2021 MSML Admissions Committee 2020-2021	Talks Lead Instructor Member Member Event Organizer Committee Member
UCSD:	Tau Beta Pi Engineering Honor Society Triton Engineering Student Council Data Science Student Society	House Leader Data Analyst Workshop Coordinator
FCC:	Google Developer Group Fresno City College Science and Engineering Club	President/Founder Treasurer